12 EXAM PRACTICE

1 The table shows the ionic radii and charges for six different ions.

Ion	J^+	L^+	M^{2+}	X^-	Y^-	Z^{2-}
Ionic radius / nm	0.14	0.18	0.15	0.14	0.18	0.15

The ionic solids JX, LY and MZ have the same lattice structure.

What is the order of magnitude of their lattice energies, giving the most exothermic first?

A JX > LY > MZ

B JX > MZ > LY

C MZ > JX > LY

D MZ > LY > JX [1]

(Total for Question 1 = 1 mark)

2 The table shows the lattice energies of rubidium fluoride, RbF, and caesium chloride, CsCl.

COMPOUND	**LATTICE ENERGY / kJ mol^{-1}**
rubidium fluoride, RbF	−783
caesium chloride, CsCl	−661

Which is most likely to be the lattice energy of caesium fluoride?

A $-647\ kJ\,mol^{-1}$ **B** $-747\ kJ\,mol^{-1}$

C $-847\ kJ\,mol^{-1}$ **D** $-947\ kJ\,mol^{-1}$ [1]

(Total for Question 2 = 1 mark)

3 Why do calcium and chlorine react together to form $CaCl_2(s)$ rather than CaCl(s)?

A Less energy is required to remove one electron from the calcium atom than to remove two electrons.

B More energy is released in forming chloride ions from chlorine molecules in the formation of $CaCl_2(s)$ than in the formation of CaCl(s).

C The lattice energy of CaCl(s) is less exothermic than the lattice energy of $CaCl_2(s)$.

D When CaCl(s) is formed from its elements, more energy is released than when $CaCl_2(s)$ is formed from its elements. [1]

(Total for Question 3 = 1 mark)

4 How does the entropy of the system change when ammonium nitrate crystals dissolve in water?

A It remains the same.

B It decreases, because the hydrated ions are more ordered in the solution than they are in the crystal.

C It increases, because the ions in the crystal become hydrated in the solution.

D It increases, because the ions are arranged more randomly in the solution than they are in the crystal. [1]

(Total for Question 4 = 1 mark)

5 Which statement is true for the exothermic reaction

$$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$$

A ΔH is positive

B $\Delta S_{surroundings}$ is positive

C ΔS_{system} is negative

D ΔS_{total} is negative [1]

(Total for Question 5 = 1 mark)

6 Calcium carbonate decomposes in an endothermic reaction when heated to 1113 K.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

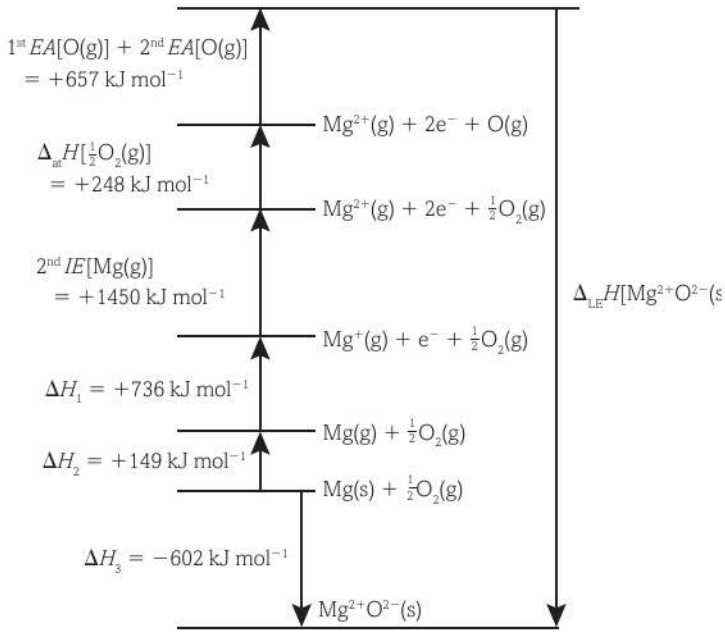
What are the signs of the entropy changes at 1113 K?

	ΔS_{system}	$\Delta S_{surroundings}$
A	+	+
B	+	−
C	−	+
D	−	−

[1]

(Total for Question 6 = 1 mark)

7 The Born–Haber cycle shown below can be used to calculate the lattice energy for magnesium oxide.



(a) Give the name of each of the enthalpy changes labelled ΔH_1, ΔH_2 and ΔH_3. [3]

(b) Write the missing formulae from the top of the Born–Haber cycle. Include state symbols. [2]

(c) The equations for the first and second electron affinities of oxygen are:

$O(g) + e^- \rightleftharpoons O^-(g)$ $\quad 1^{st}\ EA = -141\ kJ\ mol^{-1}$

$O^-(g) + e^- \rightleftharpoons O^{2-}(g)$ $\quad 2^{nd}\ EA = +798\ kJ\ mol^{-1}$

Suggest why the second electron affinity of oxygen is endothermic. [2]

(d) Use the data in the Born–Haber cycle to calculate the lattice energy of magnesium oxide. [2]

(Total for Question 7 = 9 marks)

8 Lattice energies can be found *either* via experimentally determined energy changes, using the Born–Haber cycle, *or* by theoretical calculation using the ionic model for the crystalline compound.

(a) State the energy changes that need to be determined in order to find the lattice energy of sodium fluoride, using the Born–Haber cycle. [4]

(b) The table lists the experimentally determined and theoretically calculated lattice energies for some compounds.

Compound	NaF	NaI	AgF	AgI
Experimental value/kJ mol^{-1}	−918	−705	−958	−889
Theoretical value/kJ mol^{-1}	−912	−687	−920	−778

Comment on the discrepancies between the experimental and theoretical values of the lattice energies. [4]

(c) The enthalpy changes of hydration, in kJ mol^{-1}, of the ions in the compounds listed in the table in part (b) are:

Na^+ −406 $\quad Ag^+$ −464 $\quad F^-$ −506 $\quad I^-$ −293

(i) Using the experimental lattice energy values, show by calculation which of the four compounds is the most soluble. [2]

(ii) Explain what other information is necessary in order to make a valid comparison of the solubilities of the four compounds. [2]

(Total for Question 8 = 12 marks)

9 Entropy changes are important in determining the feasibility of a process.

(a) For each of the four processes below, predict the likely sign of the entropy change of the system, ΔS_{system}. In each case, justify your answer. [4]

(i) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

(ii) $H_2O(s) \rightarrow H_2O(l)$

(iii) $2Na(s) + O_2(g) \rightarrow Na_2O_2(s)$

(iv) $S(s) + O_2(g) \rightarrow 2SO_2(g)$

(b) Magnesium oxide is used as a refractory lining for furnaces. It can be made by heating magnesium carbonate.

$MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$ $\quad \Delta H^\ominus = +117\ kJ\ mol^{-1}$

The table shows the standard entropies at 298 K of $MgCO_3(s)$, $MgO(s)$ and $CO_2(g)$.

Substance	$MgCO_3(s)$	$MgO(s)$	$CO_2(g)$
$S^\ominus$/J K^{-1} mol^{-1}	65.7	27.0	214.0

(i) Use the data to show that magnesium carbonate is stable at 298 K. [4]

(ii) Calculate the minimum temperature at which magnesium carbonate will decompose. [2]

(Total for Question 9 = 10 marks)

10 (a) The equation for the combustion of hydrogen is:

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ $\quad \Delta H^\ominus = -285.5\ kJ\ mol^{-1}$

The table gives the standard entropies at 298 K for $H_2(g)$, $O_2(g)$ and $H_2O(l)$.

Substance	$H_2(g)$	$O_2(g)$	$H_2O(l)$
$S^\ominus$/J K^{-1} mol^{-1}	131.0	205.0	69.9

(i) Calculate the standard entropy change of the system at 298 K, $\Delta S^\ominus_{system}$, for the combustion of hydrogen. [2]

(ii) Calculate the standard entropy change of the surroundings at 298 K, $\Delta S^\ominus_{surroundings}$, for the combustion of hydrogen. [2]

(iii) Calculate the total standard entropy change at 298 K, $\Delta S^\ominus_{total}$, for the combustion of one mole of hydrogen. Give your answer to an appropriate number of significant figures. [3]

(b) Explain why hydrogen does not react with oxygen unless the mixture is ignited. [1]

(Total for Question 10 = 8 marks)

11 Ammonia can be made by reacting together nitrogen and hydrogen. The reaction is reversible.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $\quad \Delta_r H^\ominus\ (700\ K) = -110.2\ kJ\ mol^{-1}$

$\Delta S^\ominus_{total}\ (700\ K) = -78.7\ J\ K^{-1}\ mol^{-1}$

(a) The $^\ominus$ symbol represents standard conditions of a specified temperature and standard pressure. In this instance, the temperature specified is 700 K. State the standard pressure for thermodynamic measurements. [1]

(b) Calculate $\Delta S^\ominus_{surroundings}\ (700\ K)$. [2]

(c) Calculate $\Delta S^\ominus_{system}\ (700\ K)$. [2]

(d) If the reaction mixture is left for long enough, in contact with a suitable catalyst, it is possible to establish an equilibrium between the nitrogen, hydrogen and the ammonia.

Comment on what the value of $\Delta S^\ominus_{total}\ (700\ K)$ indicates about the relative proportions of nitrogen, hydrogen and ammonia in the equilibrium mixture at 700 K. [1]

(Total for Question 11 = 6 marks)